Math 49 Curriculum

Precalculus with Limits | Middlesex School Mathematics | OpenStax: OpenStax Viewer

COURSE AT A GLANCE

Review Content

REVIEW

Primary: 1.2 Domain and Range; 1.5 Transformation of Functions; 4.1 Exponential Functions; 4.3 Logarithmic Functions; 4.6 Exponential and Logarithmic Equations
Supplemental: 4.2 Graphs of Exponential Functions; 4.7 Exponential and Logarithmic Models; 4.4 Graphs of Logarithmic Functions

UNIT LEARNING OBJECTIVES

I can determine the domain and range of a function from a formula, graph, or table.

I can describe and apply transformations of a parent function.

I can analyze exponential functions using their equations, graphs, and contexts.

I can analyze logarithmic functions using their equations, graphs, and contexts.

I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

Function Operations, Composition, and Inverses

REQUIRED

Primary: 1.5 Transformation of Functions; 1.4 Composition of Functions; 1.7 Inverse Functions

UNIT LEARNING OBJECTIVES

I can determine whether a function is even, odd, or neither.

I can add, subtract, multiply, and divide functions and state the resulting domain.

I can evaluate a composition from formulas, graphs, or tables.

I can write a formula for the composition of two functions and determine its domain.

I can decompose a function into a composition of simpler functions.

I can determine whether a function has an inverse on a given domain.

I can find and verify an inverse function algebraically.

I can interpret the relationship between a function and its inverse from graphs or tables.

Power Functions

REQUIRED

Primary: Precalculus 2e 3.3 Power Functions and Polynomial Functions; Precalculus 2e 3.9 Modeling Using Variation; College Algebra 2e 2.6 Other Types of Equations; Precalculus 2e 3.8 Inverses and Radical Functions
Supplemental: Precalculus 2e 3.3 Power Functions and Polynomial Functions

UNIT LEARNING OBJECTIVES

I can identify a power function and interpret the parameters in y = kx^p.

I can construct a power function from two data points or a proportional relationship.

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

I can recognize and model direct, inverse, and other power-variation relationships.

I can solve equations involving integer or rational powers.

I can build or select a power-function model for a context and interpret its parameters.

I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.

Polynomial Functions

REQUIRED

Primary: 3.4 Graphs of Polynomial Functions; 3.6 Zeros of Polynomial Functions
Supplemental: 3.6 Zeros of Polynomial Functions

UNIT LEARNING OBJECTIVES

I can predict the end behavior of a polynomial from its degree and leading coefficient.

I can determine polynomial zeros and their multiplicities.

I can construct a polynomial function from specified zeros or graphical features.

I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.

I can determine the minimum possible degree of a polynomial function from the zeros, turning points, inflection points, and end behavior of its graph.

I can build or select a polynomial model for a context and interpret its features.

Rational Functions and Limits

REQUIRED

Primary: Precalculus 2e 3.7 Rational Functions; Calculus Volume 1 2.2 The Limit of a Function; Calculus Volume 1 2.3 The Limit Laws; Precalculus 2e 3.3 Power Functions and Polynomial Functions
Supplemental: Calculus Volume 1 2.2 The Limit of a Function; Precalculus 2e 3.4 Graphs of Polynomial Functions; Precalculus 2e 3.7 Rational Functions; Precalculus 2e 4.4 Graphs of Logarithmic Functions; Precalculus 2e 3.9 Modeling Using Variation

UNIT LEARNING OBJECTIVES

I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.

I can sketch and analyze a rational function using its algebraic and asymptotic features.

I can interpret limit notation as a statement about function behavior near an input.

I can estimate a limit from a graph or table.

I can evaluate a limit algebraically using direct substitution or simplification.

I can interpret an infinite limit as vertical-asymptotic behavior.

I can use limit notation to describe behavior at infinity.

I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions, and combinations thereof.

I can build or select a rational or rational-adjacent model for a context and interpret its long-run behavior.

Sequences and Series

REQUIRED

Primary: 11.1 Sequences and Their Notations; 11.2 Arithmetic Sequences; 11.3 Geometric Sequences; 11.4 Series and Their Notations
Supplemental: 11.1 Sequences and Their Notations; 11.3 Geometric Sequences; 11.2 Arithmetic Sequences

UNIT LEARNING OBJECTIVES

I can evaluate and graph a sequence defined explicitly or recursively.

I can write explicit and recursive formulas for arithmetic sequences.

I can write explicit and recursive formulas for geometric sequences.

I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.

I can create and interpret a sequence model for discrete change.

I can interpret and use summation notation.

I can calculate a finite arithmetic series.

I can calculate a finite geometric series.

I can determine whether an infinite geometric series converges and find its sum when it does.

I can create and interpret a series model for accumulated change.

Additional Limit Concepts

EXTENSION

Primary: 2.2 The Limit of a Function; 2.4 Continuity

UNIT LEARNING OBJECTIVES

I can determine one-sided limits and decide whether a two-sided limit exists.

I can distinguish a function value from a limit and determine continuity at a point.

Parametric Equations

EXTENSION

Primary: 8.7 Parametric Equations: Graphs; 8.6 Parametric Equations

UNIT LEARNING OBJECTIVES

I can graph a plane curve defined by parametric equations with its orientation.

I can eliminate a parameter to obtain a rectangular equation.

I can create or interpret parametric equations for motion in the plane.

Conic Sections

EXTENSION

Primary: 10.1 The Ellipse; 10.2 The Hyperbola; 10.3 The Parabola
Supplemental: 10.2 The Hyperbola; 10.3 The Parabola; 8.6 Parametric Equations; 10.1 The Ellipse

UNIT LEARNING OBJECTIVES

I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.

I can use completing the square to rewrite a general-form conic equation in standard form.

I can write, interpret, and graph implicit and parametric equations of a circle.

I can write, interpret, and graph implicit and parametric equations of an ellipse.

I can write, interpret, and graph implicit and parametric equations of a hyperbola.

I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.

Conic Sections (Continued)

EXTENSION

UNIT LEARNING OBJECTIVES - CONTINUED

I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.

I can explain a parabola as the set of points equidistant from a focus and directrix.